The Impossibility of Wrapper Defense Against Prompt Injection

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Abstract

Wrapper defenses attempt to neutralize prompt injection by preprocessing inputs with a map $D\colon X \to X$. We identify a **topological obstruction** to this program: in any connected prompt space, the safe sub-level set $S_\tau = \{f < \tau\}$ is open but not closed, whereas any continuous utility-preserving defense has a closed fixed-point set containing S_τ; closure must spill onto the boundary, forcing boundary fixation. The obstruction is structural, not a matter of design or implementation. Under Lipschitz and transversality conditions, the failure is not merely existential but **quantitative**: we give explicit cone-measure and volume lower bounds on the persistent-unsafe region (Theorems E.1 and E.2), and a dilemma characterizing the optimal defense Lipschitz constant $K^* = G/\ell - 1$ (Theorem E.3). The same obstruction reappears in three extensions: discrete finite-set defenses must collapse distinct inputs to be complete; multi-turn wrappers inherit boundary fixation at every turn; stochastic wrappers with continuous expected score inherit threshold-level failure in expectation. The theory is **formalized in Lean 4 with Mathlib** across 46 files containing $\sim$360 declarations, with no admitted proofs and only three standard axioms (`https://anonymous.4open.science/r/defense-trilemma-XXXX`). The artifact's primary value is definitional scaffolding — unambiguously pinning down continuity, utility preservation, and Lipschitz regularity in a shared language that survives translation between prose and code — plus verified cross-references between theorem statements. Empirically, on real LLM safety surfaces the predicted boundary and persistence structure appears under continuous defenses, and a discrete forced-collapse experiment exhibits the information-loss mechanism predicted by the discrete theorem. Our results do not preclude training-time alignment, architectural changes, discontinuous filters, or defenses that explicitly sacrifice utility.

1 Introduction

Can you build a wrapper around a language model that eliminates all prompt injection vulnerabilities? Most wrapper-defense work implicitly assumes yes. Input classifiers that flag suspicious prompts [1], constitutional rewriting pipelines [3], and input sanitization layers [7] all share the same structure: a function $D\colon X \to X$ that preprocesses prompts before the model sees them, mapping unsafe inputs to safer equivalents while preserving benign behavior. We prove that this program has a structural failure mode. If a wrapper defense is continuous and utility-preserving, it cannot also be complete. These three properties form a **defense trilemma**: continuity, utility preservation, and completeness cannot coexist (Figure 2).

The contribution is not four unrelated theorems. It is one claim shown to survive across four regimes that matter for how wrapper defenses are used and defended in practice:

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Continuous wrapper defenses. In connected prompt spaces, a utility-preserving continuous wrapper must leave threshold-level prompts unchanged. The same obstruction survives under score-preserving and ε-relaxed utility preservation, and under Lipschitz regularity it strengthens to quantitative near-threshold and persistent-unsafe-region results (Theorems 3.1, 3.4, 3.5, 3.8 and 3.11).

Discrete wrapper defenses. Without topology, the impossibility becomes combinatorial: a complete utility-preserving defense must be non-injective. Information loss is not an implementation artifact; it is the mechanism of complete discrete defense (Theorem 4.1).

Multi-turn wrapper defenses. Interaction does not remove the obstruction. If each turn is modeled as a utility-preserving wrapper step, the same threshold-level failure reappears at every turn (Theorem 5.1).

Stochastic wrapper defenses. Randomization does not remove the obstruction either. If the expected post-defense score varies continuously and agrees with the original score on safe inputs, the stochastic defense still inherits the same boundary failure (Theorem 5.2).

Empirical validation. We then test whether the theory matters empirically. On real LLM safety surfaces, the continuous theory's boundary and persistence structure is observable; and on a saturated unsafe prompt archive, the discrete theorem's forced-collapse mechanism is directly visible.

Scope. Our results apply to *continuous, utility-preserving wrapper defenses* on *connected* spaces. They do *not* constrain training-time alignment, architectural changes, hard blocklists, output filters, ensemble/human-in-the-loop systems, or adaptive-threshold designs. All three conditions are individually necessary (Section I).

Contributions.

1. **Continuous impossibility of wrapper defense** (Theorems 3.1, 3.4, 3.5, 3.8 and 3.11): in connected prompt spaces, continuous utility-preserving wrappers must fail at the boundary; the result survives relaxed utility preservation and strengthens to quantitative and persistent-failure conclusions under additional regularity.
2. **Discrete impossibility of wrapper defense** (Theorem 4.1): on finite sets, complete utility-preserving defense necessarily collapses distinct inputs. Injective defense is necessarily incomplete.
3. **Multi-turn and stochastic impossibility of wrapper defense** (Theorems 5.1 and 5.2): the obstruction survives interaction over time and survives randomization under continuity of the induced expected score.
4. **Empirical validation**: a continuous-defense validator exhibits the predicted boundary and persistence structure on real LLM safety surfaces, and a forced-collapse experiment exhibits the discrete theorem's information-loss mechanism.
5. **Formal verification**: the full theory is verified in Lean 4 across 46 files with $\sim$360 theorems, no admitted proofs, and three standard axioms. Supporting results, full proofs, and additional empirical analyses appear in the appendix.

Related work. Adversarial robustness [20, 6, 9, 14, 21], certified defenses [5, 8, 19], and LLM jailbreaking [23, 4, 10, 12, 15, 18] provide context. Our impossibility is *map-level*: it constrains the defense function itself, not a classifier on samples or an optimizer across problems. The closest precedent is the no-free-lunch framework [22]. Full discussion in Section A.

2 Problem Setup

Definition 2.1 (Alignment Deviation Function)**.** *Let X be a topological space and let $f\colon X \to \mathbb{R}$ be a continuous scoring function. We interpret $f(x)$ as an alignment deviation score for prompt x. Given a threshold $\tau \in \mathbb{R}$:*

$$S_\tau = \{x \in X : f(x) < \tau\} \qquad \textit{(safe region)} \tag{1}$$

$$U_\tau = \{x \in X : f(x) > \tau\} \qquad \textit{(unsafe region)} \tag{2}$$

$$B_\tau = \{x \in X : f(x) = \tau\} \qquad \textit{(boundary)} \tag{3}$$

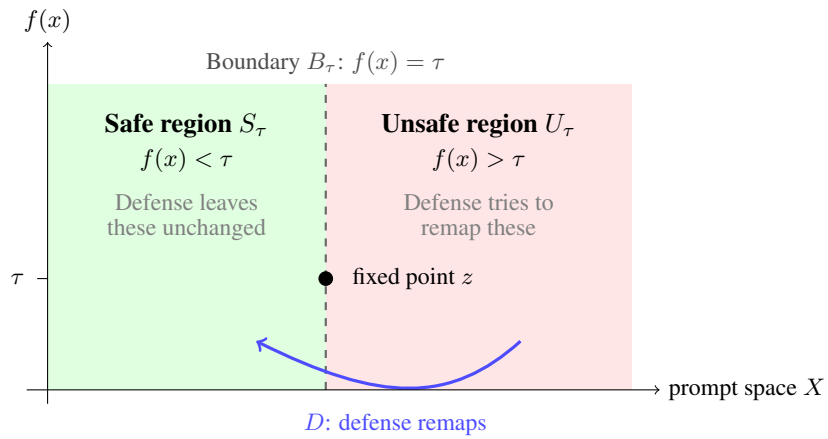

Figure 1: Schematic of the prompt space. The defense D must leave all safe inputs unchanged (utility preservation) and tries to remap unsafe inputs into the safe region. Boundary fixation (Theorem 3.1) proves the defense must also leave some boundary points unchanged, the black dot z passes through with no remediation.

Definition 2.2 (Defense)**.** *A* defense *is a continuous map* $D\colon X \to X$*. It is* utility-preserving *if* $D(x) = x$ *for all* $x \in S_\tau$*, and* complete *if* $f(D(x)) < \tau$ *for all* $x \in X$*.*

We will also use two weaker preservation notions. A defense is *score-preserving on safe inputs* if $f(D(x)) = f(x)$ for all $x \in S_\tau$, and *ε-relaxed on safe inputs* if $|f(D(x)) - f(x)| \leq \varepsilon$ for all $x \in S_\tau$.

The central question is whether a wrapper can preserve utility in one of these senses and still make all outputs safe. The following sections show that across several regimes, the answer is no.

3 Impossibility of Wrapper Defense in Continuous Settings

We begin with the most fundamental result. The argument fits in a paragraph: the defense fixes every safe input, so continuity makes its fixed-point set closed. But the safe region S_τ is open (preimage of an open interval under continuous f), and in a connected space a nonempty proper open set is not closed. Hence the fixed-point set cannot stop exactly at the edge of the safe region, it must spill onto the boundary B_τ. Some boundary prompts pass through unchanged.

Theorem 3.1 (Boundary Fixation)**.** *Let X be a connected Hausdorff space (a space that is "in one piece" and where distinct points can be separated by neighborhoods). Let $f\colon X \to \mathbb{R}$ be continuous with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ continuous with $D|_{S_\tau} = \mathrm{id}$. Then there exists $z \in X$ with $f(z) = \tau$ and $D(z) = z$. Moreover, every $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ and $D(z) = z$, and this set is nonempty.*

Proof sketch. In a Hausdorff space, $\mathrm{Fix}(D)$ is closed (preimage of the diagonal). By utility preservation, $S_\tau \subseteq \mathrm{Fix}(D)$, so $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$. But $S_\tau = f^{-1}((-\infty, \tau))$ is open and not closed (connectedness: a nonempty proper clopen set would disconnect X). Hence $\overline{S_\tau} \supsetneq S_\tau$. Any $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ (limits of values $<\tau$ cannot exceed τ, and $z \notin S_\tau$ forces $f(z) \geq \tau$) and $D(z) = z$. Full proof in Section H. □

This means that the defense's fixed-point set is too large to avoid the boundary. Utility preservation forces it to contain the safe region; closure forces it to contain the boundary; connectedness ensures the boundary is nonempty. Alternatively: a complete utility-preserving defense would be a continuous retraction $D\colon X \to S_\tau$ (since $D|_{S_\tau} = \mathrm{id}$ and $D(X) \subseteq S_\tau$), but a retract of a Hausdorff space is closed (the fixed-point set is closed), while S_τ is open and not closed (connectedness)—a contradiction.

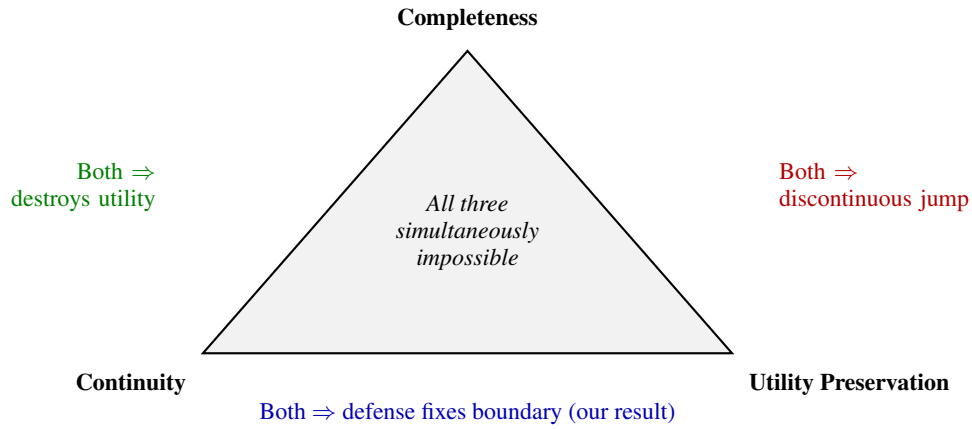

Figure 2: The defense trilemma. Any continuous wrapper defense on a connected space can satisfy at most two of the three properties. The bottom edge is our main result; the other two edges correspond to counterexamples in Section I.

Theorem 3.2 (Defense Trilemma)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. No $D\colon X \to X$ can simultaneously be continuous, utility-preserving ($D|_{S_\tau} = \mathrm{id}$), and complete ($f(D(x)) < \tau$ for all x).*

All three hypotheses are individually necessary. A defense can satisfy at most two of the three—the *defense trilemma* (Figure 2). Counterexamples for each dropped hypothesis appear in Section I.

Falsifier. An empirical counterexample to the trilemma would be a production defense that is demonstrably continuous, utility-preserving, and complete at a named τ.

Remark 3.3 (Not all boundary points are fixed)**.** *The theorem captures $\overline{S_\tau} \setminus S_\tau$, not all of B_τ. Boundary points not in $\overline{S_\tau}$ may escape fixation.*

3.1 Exact and Relaxed Utility Preservation

Strict utility preservation ($D(x) = x$ for safe x) can be relaxed. The impossibility survives score-preserving rewrites and even approximate score preservation.

Theorem 3.4 (Score-Preserving Defense)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. If $D\colon X \to X$ is continuous and* score-preserving on safe inputs*: $f(D(x)) = f(x)$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $f(D(z)) = \tau$.*

Proof sketch. Define $h = f \circ D - f$. Then h is continuous and $h|_{S_\tau} = 0$. The zero set $\{h = 0\}$ is closed and contains $\overline{S_\tau}$, since closed sets contain their boundary points. For $z \in \overline{S_\tau} \setminus S_\tau$, $f(D(z)) = f(z) = \tau$. □

The next result weakens score preservation to approximate:

Theorem 3.5 (ε-Relaxed Utility Preservation)**.** *Under the hypotheses of Theorem 3.4, if $|f(D(x)) - f(x)| \leq \varepsilon$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $f(D(z)) \geq \tau - \varepsilon$.*

Proof sketch. The set $\{x : h(x) \geq -\varepsilon\}$ is closed and contains S_τ, hence $\overline{S_\tau}$. For $z \in \overline{S_\tau} \setminus S_\tau$: $f(D(z)) = \tau + h(z) \geq \tau - \varepsilon$. □

Remark 3.6 (Why $D(S_\tau) \subseteq S_\tau$ alone is insufficient)**.** *The weakest relaxation—$D(S_\tau) \subseteq S_\tau$ with no score constraint—does allow a complete defense (e.g., a constant map $D(x) = x_0$ to a fixed safe point). But this destroys all semantic content: every prompt produces the same response. The score-preservation conditions formalize the requirement that defense must not destroy utility, without requiring the defense to be the identity.*

Remark 3.7 (Open: utility-floor relaxation)**.** *Between exact identity on S_τ and the constant map lies a realistic band of deployed wrappers that satisfy $D(S_\tau) \subseteq S_\tau$ plus a* utility floor*: e.g., embedding similarity $\mathrm{sim}(D(x), x) \geq \gamma$ for all $x \in S_\tau$. Whether the boundary-fixation obstruction persists under such a utility-floor hypothesis is an open question. A positive answer would substantially strengthen the practical reach of Theorem 3.1; a negative answer (a constructive complete defense satisfying the utility floor) would sharpen the boundary of the impossibility.*

3.2 Quantitative Failure Near the Boundary

Boundary fixation produces at least one fixed point; Lipschitz regularity makes the failure spread to a neighborhood.

Theorem 3.8 (ε-Robust Defense Constraint)**.** *Under the hypotheses of Theorem 3.1, if (X, d) is a metric space, f is L-Lipschitz, and D is K-Lipschitz, then for the fixed boundary point z:*

$$f(D(x)) \;\geq\; \tau - LK\,\mathrm{dist}(x, z) \qquad \textit{for all } x \in X. \tag{4}$$

Points within distance δ of z remain within $LK\delta$ of threshold.

Proof sketch. Since $D(z) = z$ and D is K-Lipschitz: $\mathrm{dist}(D(x), z) \leq K\,\mathrm{dist}(x, z)$. Since $f(z) = \tau$ and f is L-Lipschitz: $|f(D(x)) - \tau| \leq L \cdot K\,\mathrm{dist}(x, z)$. Full proof in Section H. □

Appendix E records explicit measure bounds for the near-threshold band and related quantitative refinements.

3.3 Persistent Unsafe Region

The ε-robust constraint bounds the depth to which the defense pushes near-boundary points, but permits values slightly below τ. When the alignment surface rises faster than the defense can pull it down, some points remain above threshold.

Decoupling global and directional Lipschitz constants. The ε-robust bound (Theorem 3.8) uses the *global* Lipschitz constant L of f, which bounds f uniformly in all directions. The persistence argument, however, compares f's growth rate in the *steep direction* to how much the defense can reduce f along the *displacement direction* $D(x) - x$. In anisotropic settings these directions differ: f may rise steeply toward the unsafe region while varying slowly in the direction the defense pulls.

We write ℓ for the *defense-path Lipschitz constant*:

$$\ell \;=\; \sup_{x \neq D(x)} \frac{|f(D(x)) - f(x)|}{\mathrm{dist}(D(x),\, x)}$$

(with $\ell = 0$ when $D = \mathrm{id}$, i.e., the supremum over the empty set is taken as 0). Since f is L-Lipschitz globally, $\ell \leq L$. When the alignment surface is **isotropic** ($\ell = L$), the steep region is empty for every $K \geq 0$ (verified in Lean as `shallow_boundary_no_persistence`). The result is non-vacuous precisely when the surface is **anisotropic**: $\ell < L$, with directional gradient G satisfying $G > \ell(K + 1)$.

Lemma 3.9 (Input-Relative Bound)**.** *Under the hypotheses of Theorem 3.8, if f has defense-path Lipschitz constant ℓ, then $f(D(x)) \geq f(x) - \ell(K + 1)\,\mathrm{dist}(x, z)$ for all $x \in X$.*

Proof sketch. Triangle inequality: $\mathrm{dist}(D(x), x) \leq \mathrm{dist}(D(x), z) + \mathrm{dist}(z, x) \leq (K+1)\,\mathrm{dist}(x, z)$. Defense-path Lipschitz: $|f(D(x)) - f(x)| \leq \ell\,\mathrm{dist}(D(x), x) \leq \ell(K + 1)\,\mathrm{dist}(x, z)$. □

This means that the defense can reduce any point's score by at most $\ell(K + 1)$ times its distance from z. If the score rises faster than that, the defense loses.

Definition 3.10 (Steep region)**.** *Given a fixed boundary point z, the* steep region *is $\mathcal{S} = \{x \in X : f(x) > \tau + \ell(K + 1)\,\mathrm{dist}(x, z)\}$—the set of points where alignment deviation exceeds τ by more than the defense's Lipschitz budget can compensate.*

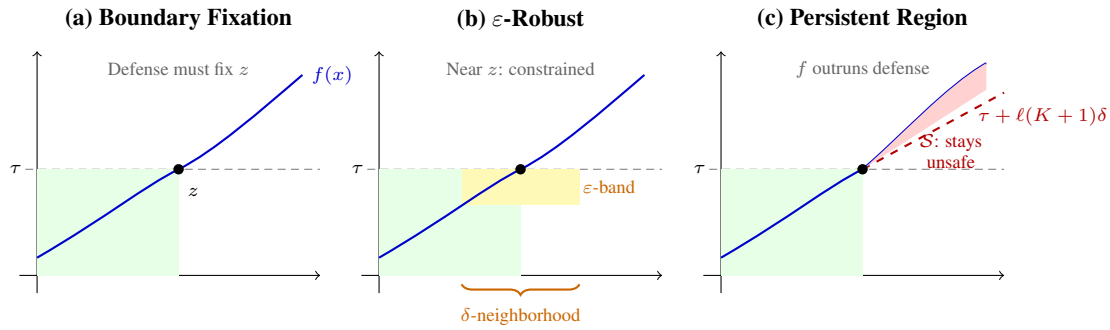

Figure 3: The three impossibility results on a 1D cross-section. **(a)** The defense must fix boundary point z (Thm. 3.1). **(b)** Near z, Lipschitz regularity constrains the defense to a shallow ε-band (yellow; Thm. 3.8). **(c)** Where f rises faster than the defense budget $\ell(K+1)\delta$ (dashed red), the region above τ persists (red shading; Thm. 3.11).

Theorem 3.11 (Persistent Unsafe Region)**.** *Let X be a connected Hausdorff metric space, f continuous and L-Lipschitz, D continuous and K-Lipschitz with $D|_{S_\tau} = \mathrm{id}$, and $S_\tau, U_\tau \neq \emptyset$. Let $z \in \overline{S_\tau} \setminus S_\tau$ be the fixed boundary point from Theorem 3.1, and let ℓ be the defense-path Lipschitz constant. If $\mathcal{S} \neq \emptyset$, then:*

1. *$\mathcal{S}$ is open.*
2. *$\mathcal{S}$ has positive measure (under any measure positive on nonempty open sets).*
3. *For every $x \in \mathcal{S}$: $f(D(x)) > \tau$.*

The defense leaves a positive-measure region that remains unsafe.

Proof sketch. $\mathcal{S}$ is a strict superlevel set of a continuous function, hence open. For $x \in \mathcal{S}$: $f(D(x)) \geq f(x) - \ell(K+1)\,\mathrm{dist}(x,z) > \tau$ by Theorem 3.9 and the definition of $\mathcal{S}$. Full proof in Section H. □

The main paper uses persistence only when the steep region is nonempty. A derivative-based sufficient condition and related gradient arguments are deferred to Sections M and Q. Empirically, the relevant models exhibit exactly this anisotropic regime (Section 6): the alignment surface rises steeply toward the unsafe region (G large) while the defense operates in a smoother subspace ($\ell \ll L$).

Falsifier. As already stated, the empirical content is $\mathrm{FP}_{\mathrm{int}} > 0$ under a continuous utility-preserving D: a cell in $\mathcal{S}_{\mathrm{pred}}$ (i.e., predicted persistent) with $f(D(x)) < \tau$ would refute Theorem 3.11.

Quantitative refinements of the continuous theory, including explicit volume lower bounds, cone-measure bounds, and the defense-aggressiveness dilemma, are deferred to the appendix (Theorems E.1 to E.3).

4 Impossibility of Wrapper Defense in Discrete Settings

On finite sets, the same failure appears as a combinatorial tradeoff: completeness vs. information preservation. No topology is required; results are verified in Lean (`MoF_12_Discrete`). The Tietze/McShane–Whitney bridge to continuous spaces is in Theorem D.1.

Theorem 4.1 (Discrete Defense Dilemma)**.** *Let X be a finite set with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ utility-preserving ($D(x) = x$ for $f(x) < \tau$).*

1. *If D is injective, then $f(D(u)) \geq \tau$ for every u with $f(u) \geq \tau$ (including boundary points): the defense is incomplete.*
2. *If D is complete ($f(D(x)) < \tau$ for all x), then D is non-injective: $\exists\, x \neq y$ with $D(x) = D(y)$.*

Proof. (1) Suppose D is injective and $f(D(u)) < \tau$ for some u with $f(u) \geq \tau$. Then $D(u)$ is safe, so $D(D(u)) = D(u)$ by utility preservation. Injectivity gives $D(u) = u$, so $f(u) = f(D(u)) < \tau$, contradicting $f(u) \geq \tau$.

(2) For any $u \in U_\tau$: completeness gives $f(D(u)) < \tau$, so utility preservation gives $D(D(u)) = D(u)$. But $u \neq D(u)$ since $f(u) \geq \tau > f(D(u))$. So u and $D(u)$ are distinct inputs with $D(u) = D(D(u))$: the defense is non-injective. □

The continuous trilemma trades *continuity*; the discrete dilemma trades *injectivity*. Any complete defense must collapse distinct inputs (Section 6).

Falsifier. A complete utility-preserving injective D on any finite input set would be a counterexample.

5 Impossibility of Wrapper Defense in Stochastic and Multi-Turn Settings

Neither repeated interaction nor randomization removes the obstruction.

5.1 Multi-Turn Impossibility

Theorem 5.1 (Multi-Turn Impossibility)**.** *Let $\{f_t, D_t\}_{t=1}^T$ be alignment functions and defenses over T turns on a connected Hausdorff space, each continuous and utility-preserving, with $S_\tau^{(t)}, U_\tau^{(t)} \neq \emptyset$ at every turn. Then for every turn t, there exists z_t with $f_t(z_t) = \tau$ and $D_t(z_t) = z_t$.*

Proof sketch. Apply Theorem 3.1 to (f_t, D_t) at each turn. The functions may depend on full history—this does not matter, as each timestep is a fresh instance of boundary fixation. □

The attacker's running-max exploit improves monotonically (`running_max_monotone`).

Falsifier. An attack trajectory with strictly decreasing running-max AD over turns would refute the monotone-improvement mechanism. Empirically, per-turn boundary fixation recurs and running-max AD increases monotonically (Table 5 and fig. 6 in the appendix).

5.2 Stochastic Defense Impossibility

Theorem 5.2 (Stochastic Defense Impossibility)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. Let D be a stochastic defense and define $g(x) = \mathbb{E}_{y\sim D(x)}[f(y)]$. If g is continuous and $g(x) = f(x)$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $g(z) = \tau$.*

Proof sketch. Define $h = g - f$. Then h is continuous and $h|_{S_\tau} = 0$, so h vanishes on $\overline{S_\tau}$ (same closure argument as Theorem 3.4). For $z \in \overline{S_\tau} \setminus S_\tau$: $g(z) = f(z) = \tau$. □

Remark (stochastic dichotomy). Since $\mathbb{E}[f(D(z))] = \tau$, either $f(D(z)) = \tau$ almost surely (the defense is deterministic at z), or the random variable $f(D(z))$ has positive probability of exceeding τ—i.e., the defense *actively produces unsafe outputs* with positive probability. The stochastic case is therefore strictly harder than the deterministic one: a genuinely random defense at boundary points must sometimes make things worse.

Remark. The continuity of g is a nontrivial assumption: it requires the distribution $D(x)$ to vary continuously with x in a suitable sense. Stochastic defenses with discontinuous rejection probabilities escape this theorem.

Falsifier. A stochastic defense at a boundary cell with $\text{variance} = 0$ and $f(D(z)) < \tau$ would refute Theorem 5.2. Empirically, a genuinely random wrapper at a boundary cell shows positive variance with positive mass above τ (Table 6 and fig. 7 in the appendix).

Pipeline composition and additional geometry/stability results are deferred to the appendix. In particular, the pipeline theorem shows that composing non-contractive stages amplifies the effective

Lipschitz constant and therefore widens the failure budget (Theorems F.1 and F.2 and sections G and J to L).

6 Experimental Validation

The empirical section tests whether the theory is vacuous. We exercise the continuous theory on real LLM safety surfaces (boundary and persistence structure) and the discrete theorem on a saturated unsafe-prompt archive (forced collapse). All experiments use the MAP-Elites behavioral-space driver of [17] over a 2D prompt space (query indirection $\times$ authority framing); every AD value comes from our own live runs (Section N).

6.1 Continuous-Setting Validation

A validator estimates L, K, ℓ, G on the grid, identifies boundary anchor z^*, and checks the containment $\mathcal{S}_{\text{pred}} \subseteq \mathcal{S}_{\text{act}}$ where $\mathcal{S}_{\text{pred}} = \{x : f(x) > \tau + \ell(K{+}1)\,\mathrm{dist}(x, z^*)\}$ and $\mathcal{S}_{\text{act}} = \{x : f(D(x)) > \tau\}$. The key metric is $\mathrm{FP}_{\mathrm{int}}$: interior cells predicted persistent but actually remediated. $\mathrm{FP}_{\mathrm{int}} > 0$ would be a real counterexample; boundary false positives merely witness discontinuity (hypothesis not met); false negatives are permitted (the theorem claims containment one way only).

We ran live against the OpenAI API with `gpt-3.5-turbo-0125` (target) and `gpt-4.1-2025-04-14` (judge). A saturation study (400 evaluations, $\sim$5,200 API calls, $\sim$\$15) filled 82 cells of a $25{\times}25$ grid with 66 unsafe ($f{>}\tau$) and 10 boundary cells—the only configuration where the theorem is non-vacuously exercised. All f values are judge-dependent; a two-judge comparison is in Section O.

6.1.1 Non-vacuous validation: continuous defenses

Theorem 3.11 has non-trivial quantitative content only when $\hat{G} > \hat{\ell}(\hat{K}{+}1)$, i.e., when the alignment surface rises faster than the defense can pull it down. A sweep over four continuous defense families (Table 10 in appendix) reveals that *most natural defenses fail this condition*: SMOOTHNEAREST SAFE, KERNELSMOOTHED, and SOFTPROJ all produce $\hat{\ell}(\hat{K}{+}1) \gg \hat{G}$ because they align displacement with $-\nabla f$, which maximizes $\hat{\ell}$. The theorem becomes a non-vacuous *diagnostic* when the defense direction diverges from ∇f: measured $(\hat{G}, \hat{\ell}, \hat{K})$ with $\hat{\ell} \ll L$ signals a construction that fights its own gradient.

We demonstrate this diagnostic on a GP-smooth oblique defense $D(x) = x + \alpha\,\beta(\mu(x))\,v(x)$ displaced 89.5° from the GP gradient (RBF kernel, $\sigma{=}0.2$, 82 cells). This is a deliberately adversarial defense direction chosen to enter the transversal regime. Estimates: $\hat{\ell} = 0.86$, $\hat{K} = 1.05$, $\hat{G} = 23.6$, so $\hat{\ell}(\hat{K}{+}1) = 1.77 \ll \hat{G}$. Result: $|\mathcal{S}_{\text{pred}}| = 3$, all actually persistent (TP=3, $\mathrm{FP}_{\mathrm{int}}$=0; Figure 4). Theorem E.3's optimal constant $K^* = \hat{G}/\hat{\ell} - 1 \approx 26.5$ gives a concrete design constraint: defenses with $K < K^*$ admit persistent cells; defenses with $K \geq K^*$ destroy utility (Figure 8 in appendix).

Appendix sweeps over three defense families, multiple kernels, and grid resolutions preserve $\mathrm{FP}_{\mathrm{int}} = 0$. Theorem 3.8's ε-robust bound holds on all 82 cells, and 10 boundary cells in $\overline{S_\tau} \setminus S_\tau$ activate boundary fixation (nearest to τ: gap $= 0.375$, vanishing under refinement). Bootstrap, seed replication, and judge-robustness checks all preserve the claim (Section N). A three-target sweep (gpt-3.5-turbo, gpt-4o-mini, gpt-4o) yields $\mathrm{FP}_{\mathrm{int}} = 0$ in every row (Table 1 in appendix).

Discrete classifier-gate footprint. Figure 5 maps two deployed gates on the same 82-cell archive. Llama-Guard-3-1B collapses to 7 MLC-taxonomy labels while passing 51/82 prompts as `safe`; ShieldGemma-2B collapses to 2 labels while passing 56/82. Both sit on the injective-incomplete horn of Theorem 4.1: structure is preserved, but majority incompleteness remains. For the forced-collapse test (Table 3 in appendix), a refusal defense collapses 66 unsafe inputs to 1 output, a canonical-category defense to 7, while an injective paraphrase defense produces 66 distinct rewrites—of which 1 remains above τ (Table 4), the predicted incomplete witness. The well-aligned target (gpt-5-mini, peak AD $= 0.5$) yields $U_\tau = \emptyset$ and the theorems are correctly silent. Full numeric tables: Tables 2 to 4.

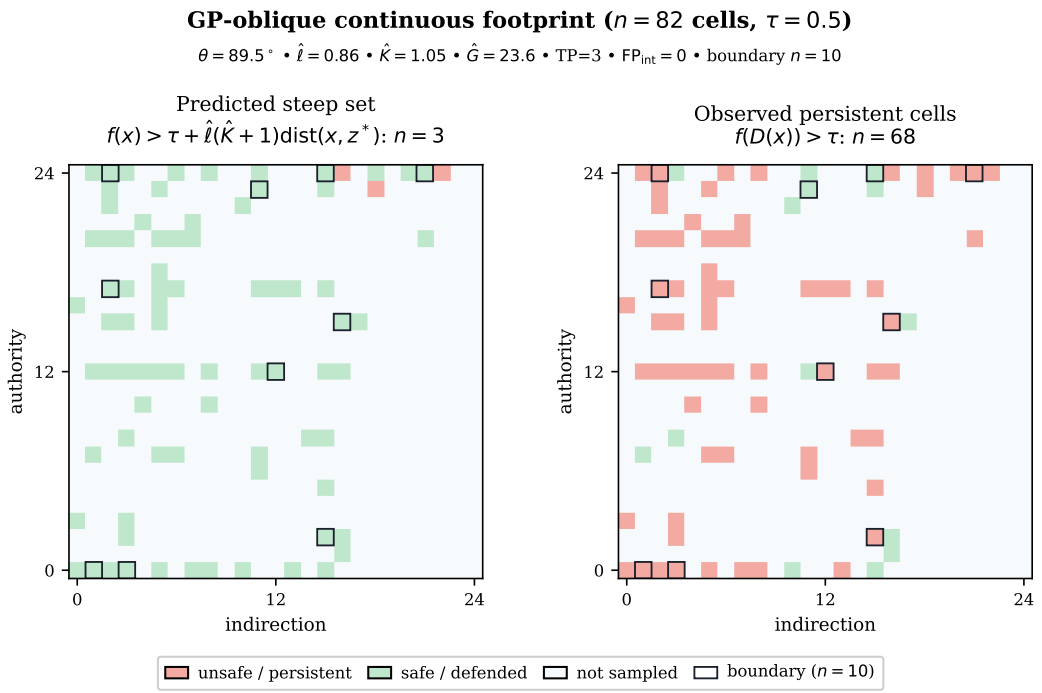

Figure 4: A *single* non-vacuous instance exercising Theorem 3.11 on the saturated `gpt-3.5-turbo-0125` grid (82 filled cells, $\tau = 0.5$) under the GP-smooth oblique defense ($\theta = 89.5°$, $\hat{\ell} = 0.86$, $\hat{K} = 1.05$). **Left:** the *steep set* $\mathcal{S}_{\text{pred}} = \{x : f(x) > \tau + \hat{\ell}(\hat{K}+1)\ \mathrm{dist}(x, z^*)\}$; $|\mathcal{S}_{\text{pred}}| = 3$. **Right:** cells actually persistent after defense, $\mathcal{S}_{\text{act}} = \{x : f(D(x)) > \tau\}$; $|\mathcal{S}_{\text{act}}| = 68$. Transversality holds non-trivially ($\hat{G} = 23.6 \gg \hat{\ell}(\hat{K}+1) = 1.77$) and the defense is genuinely non-identity (82 cells displaced). The predicted three cells are all actually persistent ($\mathrm{TP} = 3$, $\mathrm{FP}_{\mathrm{int}} = 0$). The 65 false negatives (red on the right, green on the left) are *not* theorem violations: the theorem predicts $\mathcal{S}_{\text{pred}} \subseteq \mathcal{S}_{\text{act}}$, not equality.

Scope and limitations. The theorems constrain only continuous utility-preserving wrappers on connected spaces; mechanisms outside this class (training-time alignment, architectural changes, hard blocklists, output filters, ensemble/human-in-the-loop systems) are not constrained. Persistence requires transversality ($G > \ell(K+1)$); for isotropic surfaces, $\mathcal{S} = \emptyset$. Engineering implications—including boundary shallowing, Lipschitz reduction, dimension reduction, and a worked design-constraint example—appear in Section B.

7 Conclusion

The paper makes one claim: wrapper defense is structurally incomplete under utility-preserving constraints. We show that this claim survives in four regimes. In continuous spaces, utility-preserving wrappers fix boundary prompts and, under additional regularity, leave quantitative near-threshold and persistent unsafe regions. In discrete spaces, complete defense requires input collapse. In stochastic and multi-turn settings, the same obstruction reappears rather than disappearing. Empirically, the predicted boundary and collapse mechanisms are visible in real LLM safety surfaces and prompt archives.

The practical implication is not that defense is futile. It is that wrapper defense should be engineered as risk management, not as a path to complete elimination. The right objectives are to make the boundary shallow, reduce effective dimension, monitor approach to threshold, and use mechanisms outside the wrapper model when stronger guarantees are required.

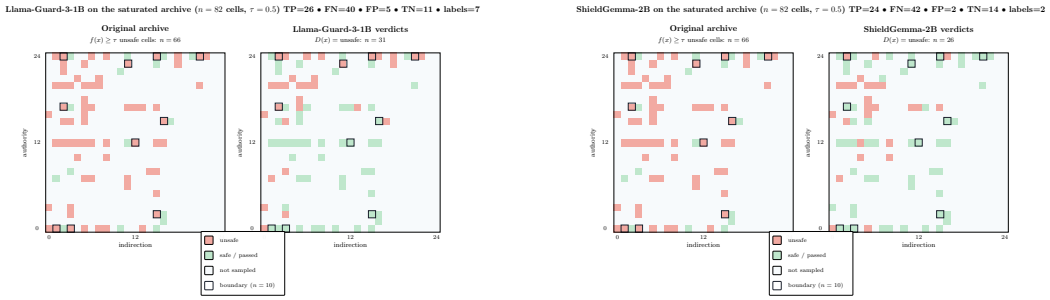

Figure 5: Behavioral-map footprint of two deployed classifier gates on the same 25×25 saturated archive. Left: original classification under the target model's AD score. Right: gate verdicts on the same filled cells. Llama-Guard-3-1B preserves 7 labels, passing 51/82 as `safe`; ShieldGemma-2B collapses to 2 labels, passing 56/82.

Compute Disclosure

The theoretical contributions of this paper—Theorems 3.1 through 5.2 and their refinements—are *Lean 4 / Mathlib* machine-verified and required no empirical compute beyond a standard `lake build` pass (minutes on a consumer laptop). All empirical compute below supports only Section 6 and Section N.

OpenAI API usage. Behavioral-manifold construction and validator runs were executed against the public OpenAI API. The aggregate API spend across all live runs referenced in the main paper and appendix—spanning the two canonical 25×25 MAP-Elites grids (`gpt-5-mini-2025-08-07` and `gpt-3.5-turbo-0125`), the saturated 400-evaluation `gpt-3.5-turbo-0125` archive ($\sim$5 200 API calls, wall-clock $\sim$130 minutes, cost $\sim$\$15), the seed-replication run (wall-clock $\sim$119 minutes), two retargeted re-evaluation surfaces (`gpt-4o-mini-2024-07-18` and `gpt-4o-2024-08-06`), a 66-prompt paraphrase pass with `gpt-4o-mini` at $T = 0$, and three judge-robustness and PAIR-baseline runs—totals approximately **\$60–\$80 USD** at list prices. Individual run wall-clocks recorded in `final_statistics.json` files for the two largest runs are `7 814 s` (saturated) and `7 139 s` (replicate); total API wall-clock across all runs is under four hours.

Local (Ollama) inference. The LlamaGuard-3-1B demonstration (Table 2) was run locally via Ollama on an Apple-silicon laptop (M-series, on-device Metal acceleration). No external compute, no GPU cluster. The full 82-prompt pass completes in under ten minutes of wall-clock.

Formal verification. Lean 4.28.0 + Mathlib v4.28.0 compiles the 46-file `ManifoldProofs` tree in approximately 20 minutes on the same consumer laptop. No GPU is used. Verification reproducibility is a function of Lean+Mathlib version pinning and is independent of API availability.

Attribution to paper sections. The \$15 saturated-archive budget supports Sections 6.1 and 6.1.1 and the entire continuous-defense sweep (Section N.3); the remaining $\sim$\$45–\$65 is divided across the retargeting sweep (Table 1), the judge-robustness deltas (Section O), seed replication (Table 14), and the PAIR baseline (Section P). The forced-collapse demonstration (Tables 3 and 4) uses the 66 paraphrases from the saturated archive and a single extra scoring pass; marginal cost below \$2.

Reproducibility envelope. All budgets quoted are order-of-magnitude; re-running the pipeline from a fresh checkout should land within a factor of 2 on cost and wall-clock, modulated by OpenAI list-price changes and model-server latency. No experiment in this paper required specialized hardware, exceeded a single laptop's memory footprint, or relied on non-public data. The anonymized artifact repository contains the exact run scripts, the raw `live_runs/` metadata (including `final_statistics.json` for each run), and the reconstruction paths from `live_runs/` to every table and figure in the paper.

Broader Impact

This paper establishes a *structural limit* on a specific class of LLM safety interventions (continuous, utility-preserving wrapper defenses). It does not propose new attacks, release new attack corpora, or provide adversaries with new techniques. We summarize the impact, dual-use, and deployment considerations below in the form encouraged by the NeurIPS 2025 Broader Impact guidance.

Responsible-disclosure stance. The artifact is a *theoretical impossibility*, not a vulnerability report. There is no affected vendor, no proof-of-concept exploit, and no coordinated-disclosure timeline to negotiate: the claim is that *any* defense satisfying continuity and utility-preservation on a connected prompt space must leave boundary prompts fixed. Responsible disclosure in this context means (i) framing the result as a design constraint rather than a recipe for bypass, (ii) providing the Lean-verified proofs so defenders can evaluate the claim on its own terms, and (iii) accompanying every empirical demonstration with explicit quantitative bounds (such as $K^* = G/\ell - 1$) that tell defenders what the achievable region looks like.

Who benefits. The primary beneficiaries are **defenders and safety researchers**: the trilemma gives them a principled way to allocate effort, a sanity check on new-wrapper proposals (if a published wrapper claims continuity, utility preservation, *and* completeness, one of the three claims must be wrong), and concrete parameter targets for the defense-design knobs they actually control (Lipschitz constant, threshold geometry, choice of feature space). Secondary beneficiaries are **auditors and policy teams** who need defensible upper bounds on what wrapper-class controls can achieve.

Who does *not* benefit. An adversary seeking to jailbreak a particular deployed model gains nothing actionable from this paper. The theorems do not identify specific prompts, do not parameterize a model family, and do not provide a search procedure over inputs. Our empirical forced-collapse demonstration (Tables 3 and 4) operates entirely on **prompts already contained in public jailbreak datasets**; no new adversarial text is produced by this work that was not already available to adversaries via prior publications and public leaderboards.

Dual-use analysis. We considered three plausible misuse vectors:

- *"The paper implies wrappers are useless, so vendors will stop deploying them."* The paper explicitly rules this out (Section B): wrappers remain the cheapest risk-reduction primitive, they just cannot be total. This is analogous to the existence of adversarial examples not implying that image classifiers should be removed from production.
- *"The paper tells attackers where to look."* The boundary-fixation phenomenon is a consequence of continuity on a connected space, not a localized exploit. No attacker gains a search advantage from knowing that a boundary exists in general.
- *"The paper's quantitative bounds help adversaries tune."* The quantitative bounds (Theorems 3.11–E.2 and $K^* = G/\ell - 1$) are defender-facing: they require the *wrapper's* Lipschitz constant and the *surface's* boundary rate, both of which are available only to the operator running the safety stack, not to an external attacker probing the deployed system.

Mitigations. To minimize misuse potential, this paper (i) does not propose or evaluate any new attack; (ii) releases only *wrapper-side* quantitative bounds (Lipschitz constant, persistent-unsafe measure, threshold gap) rather than attack improvements; (iii) anonymizes the artifact URL to comply with double-blind review and prevent author-linked attack-adjacent discoverability; and (iv) pairs every empirical table with the theorem it exercises and the inequality it is expected to satisfy, so that readers cannot mistake the demonstrations for generic red-teaming results.

Deployment guidance for practitioners. For a team deploying an LLM-backed product, the intended takeaway is a *design-constraint tool*, not an attack toolkit. Concretely: (a) if a wrapper is continuous and utility-preserving, treat its near-threshold behavior as structurally unreliable and do not rely on it alone; (b) invest the saved attention in layers outside the wrapper class—training-time alignment, output-side filters, human-in-the-loop review of boundary cases, and architectural controls on the underlying model; (c) measure $\hat{L}, \hat{\ell}, \hat{G}, \hat{K}$ on your own surface and use $K^* = \hat{G}/\hat{\ell} - 1$ as a planning number when specifying SLAs for the wrapper layer; (d) if regulatory or contractual

requirements demand *complete* wrapper safety, know that this forces either discontinuity (e.g., hard blocklists with a non-negligible refusal rate on benign inputs) or utility sacrifice (e.g., canonicalizing prompt content), and plan the product experience accordingly.

Scope of this statement. Our results do not address fairness, bias, privacy, or environmental impact of LLMs more broadly; they address a specific class of safety controls and a specific structural limit on that class. Claims about adequacy of alternative defense mechanisms (RLHF, constitutional training, discrete filtering, ensembles) are out of scope.

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Appendices

A Related Work

Adversarial robustness research has established that small perturbations can fool classifiers [20, 6, 9, 14] and that robustness may fundamentally trade off against accuracy [21, 11]. Certified defenses provide guarantees for fixed models [5, 8, 19, 13, 2], and topological perspectives have illuminated the structure of decision boundaries [16]. For LLMs specifically, jailbreaking attacks [23, 4, 10, 12] and automated red-teaming [15, 18] demonstrate persistent vulnerabilities.

Our work differs in object and claim. Rather than studying how failures arise for a fixed model or how a particular system can be certified, we impose universal constraints on the wrapper map itself. The closest conceptual precedent is the no-free-lunch framework [22]: the result is not that one defense is poorly implemented, but that a defense pattern cannot satisfy all of its intended requirements simultaneously.

Unlike no-free-lunch (which constrains optimizers across problems) or robustness-accuracy trade-offs [21] (which constrain classifier behavior on samples), our impossibility is *map-level* — it constrains the defense function itself, independent of which attacks or samples are considered. The topological obstruction S_τ open-but-not-closed applies to every continuous utility-preserving $D\colon X \to X$, not to a learning problem or to a distribution of inputs.

B Implications, Scope, and Limitations

The results do not say defense is valueless; they say *complete* defense is impossible under the stated constraints. The engineering goal shifts from elimination to management:

1. Make the boundary shallow. Set τ so that boundary-level behavior is benign. If $f(z) = \tau$ yields a polite refusal rather than harmful compliance, the impossibility is mathematically true but practically harmless. GPT-5-Mini exemplifies this: its ceiling at $\mathrm{AD} = 0.50$ means $U_\tau = \emptyset$, so the impossibility theorems do not apply.

2. Reduce the Lipschitz constant. Smaller L tightens $\ell \leq L$, potentially narrowing the persistent region.

3. Reduce the effective dimension. Under grid-based exhaustive-enumeration defenses, cost grows as N^d (Theorem L.1); learning-based defenses that generalize across the space may sidestep this exponential bound. Constraining the prompt interface reduces d regardless of defense strategy.

4. Monitor, don't eliminate, the boundary. Transversal crossings persist under fine-tuning (Theorem K.2) and recur every turn (Theorem 5.1). The Lipschitz bound gives a computable estimate of distance to boundary.

Worked example (Theorem E.3). For $\hat{\ell} = 0.86$ and $\hat{G} = 23.6$, the optimal $K^* = \hat{G}/\hat{\ell} - 1 \approx 26.5$. Designs with $K < K^*$ admit a persistent region; designs with $K \geq K^*$ destroy utility.

Boundary fixation is at the boundary. Fixed points satisfy $f(z) = \tau$ exactly. If τ-level behavior is benign, the theorem is true but harmless.

Persistence requires transversality. For isotropic f ($\ell = L$), $\mathcal{S}$ is empty for all $K \geq 0$.

Grid-based cost asymmetry. Theorem L.1 assumes exhaustive grid enumeration. Learning-based defenses may sidestep the exponential bound.

Table 1: Continuous-defense validator across three OpenAI target models. $\mathrm{FP}_{\mathrm{int}} = 0$ holds across all three.

Target	Cells	Peak AD	$\lvert\mathcal{S}_\tau\rvert$	$\lvert\mathcal{U}_\tau\rvert$	L	K	ℓ	G	$\lvert\mathcal{S}_{\mathrm{pred}}\rvert$	$\mathrm{FP}_{\mathrm{int}}$	role
`gpt-3.5-turbo-0125`	82	1.00	16	66	23.62	3.16	23.62	23.62	1	0	weak
`gpt-4o-mini-2024-07-18`	82	0.99	13	24	12.25	4.94	4.02	10.12	0	0	sil.
`gpt-4o-2024-08-06`	82	0.99	10	30	12.38	3.41	12.38	12.38	0	0	sil.

Table 2: Llama-Guard-3-1B applied to the 82 unsafe-seed prompts, demonstrating Theorem 4.1's injective-incomplete horn. 52 of 82 attacks pass as `safe`.

Quantity	Value
Input prompts $\lvert X\rvert$	82
Distinct Llama-Guard labels $\lvert D(X)\rvert$	7
Collapse rate $1 - \lvert D(X)\rvert/\lvert X\rvert$	0.915
Largest collision class	`safe` ($n = 51$)
Verdict `unsafe`	31
Verdict `safe` (false neg.)	51
Verdict other/error	0
Theorem 4.1 horn	*injective-but-incomplete*

Code	Hazard category	# prompts
S9	Indiscriminate Weapons	8
S2	Non-Violent Crimes	8
S1	Violent Crimes	9
S6	Specialized Advice	4
S10	Hate	1
S7	Privacy	1
–	safe (false negatives)	51

C Deferred Empirical Illustrations

D Continuous Interpolation of Discrete Observations

If one wishes to embed discrete behavioral observations into a continuous prompt space, the bridge is standard.

Theorem D.1 (Continuous Relaxation)**.** *Let (X, d) be a connected, normal, Hausdorff metric space, and $S \subset X$ a finite set of observations with observed alignment scores $g\colon S \to \mathbb{R}$ satisfying $g(p) < \tau$ and $g(q) > \tau$ for some $p, q \in S$. Then there exists a continuous $f\colon X \to \mathbb{R}$ extending g (i.e., $f|_S = g$) for which the hypotheses of Theorem 3.1 hold. If the extension is chosen to be Lipschitz (via McShane–Whitney), the hypotheses of Theorem 3.8 also hold. If f is additionally Lipschitz, Theorem 3.11 applies wherever transversality is met.*

Proof sketch. **Step 1** (Extension exists). S is finite and X is T_1, so S is closed. Since X is normal and $g\colon S \to \mathbb{R}$ is continuous on a discrete closed subset, the Tietze extension theorem provides a continuous $f\colon X \to \mathbb{R}$ with $f|_S = g$.

Step 2 (Hypotheses of Theorem 3.1 are satisfied). We have $f(p) = g(p) < \tau$ and $f(q) = g(q) > \tau$, so $S_\tau \neq \emptyset$ and $U_\tau \neq \emptyset$. Since X is connected and Hausdorff, Theorem 3.1 applies.

Step 3 (Lipschitz extension enables stronger results). When a Lipschitz extension is needed, the McShane–Whitney theorem provides an L-Lipschitz f agreeing with g on S. Then Theorem 3.8 and Theorem 3.11 apply under their additional hypotheses. □

E Quantitative Refinements

The main paper establishes existence and persistence. This section records explicit quantitative refinements.

Table 3: Forced-collapse demonstration of Theorem 4.1 on 66 unsafe prompts. Complete defenses collapse 66 inputs to 1 or 7 outputs; the injective paraphrase defense keeps all 66 distinct.

Defense	Inputs	Distinct outputs	Collapse rate	Max collision class
refusal	66	1	0.985	66
canonical-category	66	7	0.894	40
paraphrase	66	66	0.000	1

Table 4: Paraphrase defense is injective (66 distinct outputs), so Theorem 4.1 requires incompleteness. Rescoring confirms $1/66$ remains above τ.

Defense	$\lvert\{x : f(D(x))>\tau\}\rvert$	Incomplete per Thm. 4.1
paraphrase (T=0, strict $> \tau$)	$1/66$	✓
paraphrase (T=0, borderline $\geq \tau$)	$5/66$	✓

Theorem E.1 (Volume Lower Bound)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be L-Lipschitz with $L > 0$, and let μ denote Lebesgue measure. If there exists c with $f(c) = \tau - \varepsilon/2$, then $B(c, \varepsilon/(4L)) \subseteq \mathcal{B}_\varepsilon$, giving*

$$\mu(\mathcal{B}_\varepsilon) \;\geq\; V_n \cdot \left(\frac{\varepsilon}{4L}\right)^n,$$

where V_n is the volume of the unit ball in $\mathbb{R}^n$. In $\mathbb{R}^1$, this simplifies to $\mu(\mathcal{B}_\varepsilon) \geq \varepsilon/(2L)$.

Proved in the Lean artifact as `MoF_17_CoareaBound`.

Theorem E.2 (Cone Measure Bound)**.** *In $\mathbb{R}$ with Lebesgue measure μ, if $f(x) \geq \tau + c(x - z)$ for all $x \in (z,\, z + \delta_0)$ with $c > \ell(K + 1)$, then $\mu(\{x : f(D(x)) > \tau\}) \geq \delta_0$.*

Proved in the Lean artifact as `MoF_18_ConeBound`.

Theorem E.3 (Defense Dilemma)**.** *Assume f is differentiable at boundary point z with $G = \|\nabla f(z)\|$, and let ℓ be the defense-path Lipschitz constant. Define $K^* = G/\ell - 1$. Then:*

1. *If $K < K^*$, the persistent unsafe region exists ($G > \ell(K + 1)$, so Theorem 3.11 applies).*
2. *If $K \geq K^*$, the ε-robust bound becomes too loose to exclude success on the steep region.*

Since $\ell \leq L$, the dilemma is sharpest when $\ell \ll L$.

Proved in the Lean artifact as `MoF_19_OptimalDefense`.

F Pipeline Composition

Theorem F.1 (Pipeline Lipschitz Degradation)**.** *If stages $T_1, \ldots, T_n$ are $K_1, \ldots, K_n$-Lipschitz, the composed pipeline is $(\prod K_i)$-Lipschitz. For n stages each with $K \geq 2$, the effective constant is K^n.*

Falsifier. A measured pipeline with empirical composed Lipschitz constant $\hat{K}_{\text{composed}} > \hat{K}_1 \cdot \hat{K}_2$ would refute the bound in Theorem F.1: Lipschitz constants submultiply under composition, so the composed constant can never exceed the product.

Theorem F.2 (Pipeline Impossibility)**.** *If the composed pipeline $P = T_n \circ \cdots \circ T_1 \circ D$ is continuous and $P(x) = x$ for all $x \in S_\tau$, then P has boundary fixed points. If D is K_D-Lipschitz and each T_i is K-Lipschitz, the ε-robust band scales as $L \cdot K_D \cdot K^n \cdot \delta$.*

Proved in the Lean artifact as `MoF_15_NonlinearAgents`.

G Vulnerability Landscape

This section characterizes the geometry of the unsafe region.

Theorem G.1 (Basin Structure)**.** *If f is continuous and $f(p) > \tau$, then U_τ is open. Under any measure positive on nonempty open sets, U_τ has positive measure.*

Table 5: Per-turn boundary-fixation statistics. Running-max AD is monotone non-decreasing.

Traj	AD_1	AD_2	AD_3	AD_4	AD_5	$\max_k$
T1	0.500	0.500	0.500	0.500	0.900	**0.900**
T2	0.990	0.500	0.500	0.500	0.500	**0.990**
T3	1.000	1.000	1.000	0.995	1.000	**1.000**
T4	0.990	0.500	0.500	0.500	0.500	**0.990**
T5	1.000	1.000	1.000	0.990	0.990	**1.000**
T6	1.000	0.500	0.500	0.500	0.500	**1.000**
T7	0.990	0.990	1.000	0.990	1.000	**1.000**
T8	0.975	0.500	0.500	0.500	0.900	**0.975**
T9	1.000	1.000	1.000	0.500	0.990	**1.000**
T10	0.500	0.975	0.500	0.500	0.500	**0.975**

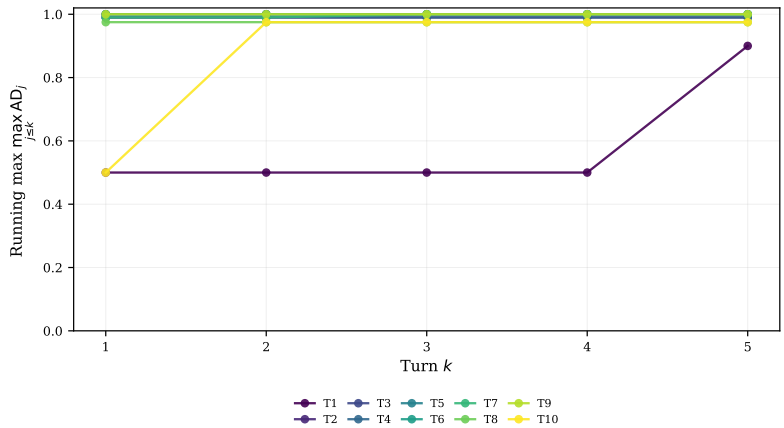

Figure 6: Multi-turn running-max AD trajectory; boundary-fixed points recur at each turn.

Theorem G.2 (Basin Fragment Minimum Size)**.** *If X is a normed space, f is L-Lipschitz with $f(p) > \tau$, the connected component of U_τ containing p has diameter $\geq 2(f(p) - \tau)/L$.*

Smoother surfaces (smaller L) produce larger basins; rougher surfaces produce smaller fragments.

H Full Proofs

Proof of Theorem 3.1 (Boundary Fixation). **Step 1** (Hausdorff $\Rightarrow$ fixed-point set is closed). $\mathrm{Fix}(D) = \{x : D(x) = x\}$ is the preimage of the diagonal $\Delta \subset X \times X$ under $x \mapsto (D(x), x)$. In a Hausdorff space, Δ is closed, so $\mathrm{Fix}(D)$ is closed.

Step 2 (Utility preservation $\Rightarrow$ safe region $\subseteq$ fixed points). $S_\tau \subseteq \mathrm{Fix}(D)$. Since $\mathrm{Fix}(D)$ is closed: $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$.

Step 3 (Connectedness $\Rightarrow$ safe region is not closed). $S_\tau = f^{-1}((-\infty, \tau))$ is open. If also closed, it would be clopen—but in a connected space the only clopen sets are $\emptyset$ and X. Since both S_τ and U_τ are nonempty, S_τ is not closed.

Step 4 (Boundary point exists). $\overline{S_\tau} \supsetneq S_\tau$, so there exists $z \in \overline{S_\tau} \setminus S_\tau$. Continuity gives $f(z) \leq \tau$; $z \notin S_\tau$ gives $f(z) \geq \tau$. Hence $f(z) = \tau$.

Step 5 (Defense fixes the boundary point). $z \in \overline{S_\tau} \subseteq \mathrm{Fix}(D)$, so $D(z) = z$ and $f(D(z)) = \tau$. $\square$

Proof of Theorem 3.8 (ε-Robust Constraint). By Theorem 3.1, the fixed boundary point z exists.

Step 1. Since $D(z) = z$ and D is K-Lipschitz: $\mathrm{dist}(D(x), z) = \mathrm{dist}(D(x), D(z)) \leq K\,\mathrm{dist}(x, z)$.

Step 2. Since $f(z) = \tau$ and f is L-Lipschitz: $|f(D(x)) - \tau| = |f(D(x)) - f(z)| \leq L\,\mathrm{dist}(D(x), z) \leq LK\,\mathrm{dist}(x, z)$. $\square$

Table 6: Stochastic-defense statistics at a boundary cell. Positive variance forces positive mass above τ.

Boundary cell	original AD	mean $f(D(x))$	std	max	$P(f(D(x))>\tau)$
$[4, 21]$	0.850	0.513	0.072	0.900	0.03 (1/30)
$[14, 8]$	0.850	0.562	0.142	0.890	0.17 (5/30)
$[1, 24]$	0.875	0.526	0.100	0.900	0.07 (2/30)
$[3, 0]$	0.875	0.526	0.100	0.900	0.07 (2/30)
$[2, 15]$	0.890	0.500	0.000	0.500	0.00 (0/30)

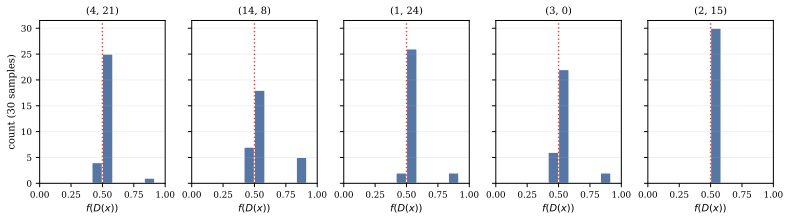

Figure 7: Histogram of $f(D(z))$ at a boundary cell under a stochastic wrapper; $\mathbb{E}[f(D(z))] = \tau$ and positive variance pushes mass above τ.

Proof of Theorem 3.11 (Persistent Unsafe Region). **(1)** $\mathcal{S}$ is the strict superlevel set of $x \mapsto f(x) - \ell(K+1)\,\mathrm{dist}(x, z)$ at level τ, hence open.

(2) Open and nonempty implies positive measure.

(3) For $x \in \mathcal{S}$: $f(x) > \tau + \ell(K+1)\,\mathrm{dist}(x, z)$. By Theorem 3.9: $f(D(x)) \geq f(x) - \ell(K+1)\,\mathrm{dist}(x, z) > \tau$.

Lean: `persistent_unsafe_refined` *in* `MoF_20_RefinedPersistence` *(defense-path constant ℓ).* `MoF_11_EpsilonRobust` *contains an earlier version using the global constant L; as noted in Section 3.3, that version is vacuous for isotropic surfaces.* □

I Counterexamples: Each Hypothesis Is Necessary

Counterexample I.1 (Removing connectedness)**.** *$X = \{0, 1\}$ discrete, $f(0) = 0$, $f(1) = 1$, $\tau = 0.5$. $D(0) = 0$, $D(1) = 0$: continuous, utility-preserving, complete.*

Counterexample I.2 (Removing continuity)**.** *$X = [0, 1]$, $f(x) = x$, $\tau = 0.5$. $D(x) = x$ for $x < 0.5$, $D(x) = 0$ for $x \geq 0.5$: utility-preserving and complete, but discontinuous at 0.5.*

Counterexample I.3 (Removing utility preservation)**.** *$D(x) = x_0$ for a fixed safe point: continuous and complete, but destroys all inputs.*

J Attack Properties

Theorem J.1 (Perturbation Robustness)**.** *If f is L-Lipschitz and $f(p) > \tau$, then $B(p, (f(p) - \tau)/L) \subseteq U_\tau$. The radius is monotone in $f(p)$.*

Theorem J.2 (Iterative Convergence)**.** *Any monotone-improvement operator T with score in $[0, 1]$ converges. If each step gains $\geq \delta$, convergence takes $\leq \lfloor 1/\delta \rfloor$ steps.*

Theorem J.3 (Transferability)**.** $\|f - g\|_\infty \leq \delta \implies \{f > \tau + \delta\} \subseteq \{g > \tau\}$. *Transfer costs zero additional queries.*

Theorem J.4 (Authority Monotonicity)**.** *If $f(y, \cdot)$ is monotone non-decreasing in authority for each fixed indirection y, the vulnerability set is upward-closed with a critical threshold $a_2^*(y)$ (by IVT, when $f(y, \cdot)$ is continuous and crosses τ). If additionally f is monotone non-decreasing in indirection for each fixed authority, the threshold curve $y \mapsto a_2^*(y)$ is non-increasing.*

Theorem J.5 (Gradient Ascent)**.** *If f has nonzero Fréchet derivative at x, there exist v and $\varepsilon > 0$ with $f(x + \varepsilon v) > f(x)$.*

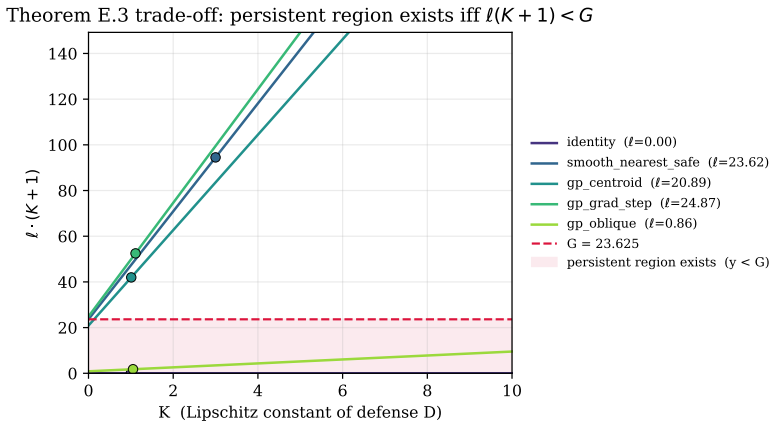

Figure 8: Persistence-vs-utility tradeoff as K varies around $K^* = G/\ell - 1$ from Theorem E.3. Below K^* the persistent unsafe region is nonempty; at and above K^* the displacement needed to kill persistence is incompatible with utility preservation.

Table 7: Empirical pipeline Lipschitz degradation. The measured composed constant $\hat{K}_{\text{composed}}$ is bounded above by $\prod \hat{K}_i$, consistent with Theorem F.1.

stage 1	stage 2	K_1	K_2	K_1K_2	K_{comp}	$K_{\text{comp}} \leq K_1K_2$?
`smooth-nearest-safe`	`kernel-smoothed`	3.00	1.71	5.13	5.10	✓
`smooth-nearest-safe`	`soft-projection`	3.00	7.07	21.21	10.44	✓
`kernel-smoothed`	`smooth-nearest-safe`	1.71	3.00	5.13	3.48	✓
`kernel-smoothed`	`soft-projection`	1.71	7.07	12.08	4.47	✓

K Stability Under Fine-Tuning

Theorem K.1 (Interior Stability)**.** *If $\|f - g\|_\infty \leq \varepsilon$: $f(x) > \tau + \varepsilon \implies g(x) > \tau$, and $f(x) < \tau - \varepsilon \implies g(x) < \tau$. Only the band $|f(x) - \tau| \leq \varepsilon$ is uncertain.*

Theorem K.2 (Crossing Preservation)**.** *Let $f, g\colon [a, b] \to \mathbb{R}$ be continuous. If f crosses τ on $[a, b]$ with margin m (i.e., $f(a') < \tau - m$ and $f(b') > \tau + m$ for some $a', b' \in [a, b]$) and $\|f - g\|_\infty \leq \varepsilon < m$, then g also crosses τ on $[a, b]$.*

Theorem K.3 (Patching Is Nonlocal)**.** *There exist f, g with $\|f - g\|_\infty \leq \varepsilon$ such that eliminating a vulnerability at one point necessarily changes values at distant points.*

L Cost Asymmetry

Theorem L.1 (Cost Asymmetry for Grid-Enumeration Defenses)**.** *For* grid-based exhaustive-enumeration *defense with $N \geq 2$ bins per axis in d dimensions: attack cost $\leq 1/\delta$ (dimension-independent); defense cost $= N^d$ (exponential in d); ratio $\delta \cdot N^d \to \infty$ as $d \to \infty$.*

Caveat. This bound applies to grid enumeration, not to learning-based defenses (e.g., GP surrogates) that generalize across the space without evaluating every cell. The paper's own empirical section uses such surrogates at cost $\ll N^d$.

At $d = 2$ with $N = 25$ and $\delta = 0.01$, the ratio is 6.25. At $d = 10$, it climbs to $\sim 10^{12}$.

M Additional Verified Results

Lipschitz displacement bound. D is K-Lipschitz with $D(z) = z \implies \operatorname{dist}(D(x), x) \leq (K + 1) \operatorname{dist}(x, z)$.

Table 8: Counterexample ablation. Each row drops one of the three hypotheses and reports whether the defense becomes complete, together with the price paid in the other two properties.

Case	Hypothesis dropped	Construction	Validator flag
C.1	connectedness	$X = \{0, 1\}$ discrete, $D = \mathrm{id}$	`applicable=False`; $\overline{S_\tau} = S_\tau$
C.2	continuity	$X = [0, 1]$, $D(x) = x \cdot \mathbf{1}_{x<0.5}$	local Lipschitz $K = 49.0$ at x=0.495
C.3	utility preservation	$D(x) = x_0$ constant	$\lvert D(X)\rvert = 1$; 50/50 safe inputs displaced

Tool calls amplify failure. Each non-contractive tool call multiplicatively increases the pipeline's effective Lipschitz constant: $K^n < K^{n+1}$ for $K \geq 2$.

Attacker monotone improvement. Best observed alignment deviation is non-decreasing across turns.

Attacker steering. If directional slope varies continuously with attacker parameter α and crosses $\ell(K + 1)$, transversality is reachable by IVT.

Stochastic regularity. $g(x) = \mathbb{E}[f(D(x))]$ satisfies $g - f = 0$ on $\overline{S_\tau}$.

Discrete defense dilemma. An injective, utility-preserving defense is incomplete (every unsafe input stays unsafe). A complete, utility-preserving defense is non-injective (distinct inputs collapse to the same output). The three properties—completeness, utility preservation, injectivity—form a trilemma.

Defense position invariance. Lipschitz constant of defense-before-tools equals defense-after-tools: $K_D \cdot K_T^n$.

N Additional Experimental Details

This appendix collects the estimator definitions, sanity checks, continuous-defense sweeps, and robustness tables that support the main paper's empirical claims.

N.1 Estimator definitions

Let G_f denote the set of filled cells in an $N \times N$ MAP-Elites grid and let $h = 1/N$ be the cell width. Distances use normalized L_2 grid coordinates on $[0, 1]^2$:

$$\mathrm{dist}\big((i_1, j_1), (i_2, j_2)\big) = h\sqrt{(i_1 - i_2)^2 + (j_1 - j_2)^2}.$$

All pairs involving unfilled cells are excluded; no imputation is performed. The boundary anchor is

$$z^* = \arg\min_{x \in G_f} |f(x) - \tau|,$$

restricted to sampled boundary cells $\overline{S_\tau} \setminus S_\tau$, with lexicographic tie-breaking.

Surface Lipschitz constant.

$$\hat{L} = \max_{x,y \in G_f,\ x \neq y} \frac{|f(x) - f(y)|}{\mathrm{dist}(x, y)}.$$

Defense Lipschitz constant.

$$\hat{K} = \max_{u,v \in G_f,\ u \neq v} \frac{\mathrm{dist}(D(u), D(v))}{\mathrm{dist}(u, v)}.$$

Defense-path Lipschitz constant.

$$\hat{\ell} = \max_{x \in G_f,\ D(x) \neq x,\ D(x) \in G_f} \frac{|f(D(x)) - f(x)|}{\mathrm{dist}(D(x), x)}.$$

If the set is empty, we take $\hat{\ell} = 0$.

Boundary directional rate.

$$\hat{G} = \max_{x,y \in G_f,\ f(x)<\tau<f(y)} \frac{f(y) - f(x)}{\mathrm{dist}(x, y)}.$$

These are the estimators used by the validator and the tables below.

N.2 Identity and out-of-scope sanity checks

Identity-defense rows are bookkeeping checks rather than theorem tests: under $D = \mathrm{id}$, the predicted and observed unsafe sets coincide by construction.

Table 9: **Smoke tests (not a test of the theorem).** The two sparse 25×25 live runs under the canonical identity defense. Under $D = \mathrm{id}$, $\mathcal{S}_{\text{pred}} \equiv \mathcal{S}_{\text{act}} \equiv \{x : f(x) > \tau\}$ by construction, so agreement only checks the validator's bookkeeping.

Target	$\boldsymbol{\tau}$	**Filled**	$\boldsymbol{L}$	$\boldsymbol{G}$	$\lvert\mathcal{S}_{\text{pred}}\rvert$	$\lvert\mathcal{S}_{\text{act}}\rvert$	**TP**	**FP**$_{\text{int}}$	**FN**
gpt-5-mini-2025-08-07	0.30	14	6.13	6.13	7	7	7	**0**	0
gpt-3.5-turbo-0125 (smoke)	0.50	9	11.44	11.44	7	7	7	**0**	0

N.3 Continuous-defense sweeps

Table 10: Continuous-defense sweep on the saturated `gpt-3.5-turbo-0125` grid over continuous defense families (GP-oblique, Gaussian-smoothed nearest-safe, contractive-flow). Each row reports empirical $\hat{\ell}$, $\hat{K}$, $|\mathcal{S}_{\text{pred}}|$, $|\mathcal{S}_{\text{act}}|$, TP, and FP$_{\text{int}}$.

defense	L	K	$\hat{\ell}_{\text{diag}}$	$\hat{\ell}_{\text{diag}}(K{+}1)$	G	trans.	$\lvert\mathcal{S}_{\text{pred}}\rvert$	TP	FP$_{\text{int}}$	$\lvert\mathcal{S}_{\text{act}}\rvert$	role
SMOOTHNEARESTSAFE	23.62	3.162	23.625	98.334	23.62	×	1	1	0	59	weak
KERNELSMOOTHED	23.62	5.000	21.750	130.500	23.62	×	1	1	0	39	weak
SOFTPROJ	23.62	2.000	23.625	70.875	23.62	×	1	0	0	56	mix.
OBLIQUEGP	23.62	1.050	0.274	0.563	23.62	✓	46	46	0	68	info.

Table 11: GP-kernel sensitivity for the oblique-smooth defense on the saturated `gpt-3.5-turbo-0125` grid. Kernel $\in \{\text{RBF}, \text{Matérn-}3/2, \text{Matérn-}5/2\}$ crossed with bandwidth $\sigma \in \{0.1, 0.2, 0.4\}$.

kernel	σ	L	K	$\hat{\ell}_{\text{diag}}$	G	$\lvert\mathcal{S}_{\text{pred}}\rvert$	TP	FP$_{\text{int}}$
RBF	0.10	23.62	1.105	0.884	23.62	9	9	0
RBF	0.20	23.62	1.050	0.274	23.62	46	46	0
RBF	0.40	23.62	1.037	0.099	23.62	71	71	0
Matérn-3/2	0.10	23.62	1.068	1.315	23.62	7	7	0
Matérn-3/2	0.20	23.62	1.070	1.165	23.62	8	8	0
Matérn-3/2	0.40	23.62	1.070	0.946	23.62	8	8	0
Matérn-5/2	0.10	23.62	1.079	0.899	23.62	9	9	0
Matérn-5/2	0.20	23.62	1.080	0.734	23.62	9	9	0
Matérn-5/2	0.40	23.62	1.094	0.345	23.62	41	41	0

N.4 Robustness checks

N.5 Out-of-scope discrete stress tests

Bounded-step defenses are not continuous, so Theorem 3.11's hypotheses do not apply. We include them only as negative consistency checks.

Table 12: Resolution sweep for the oblique-smooth defense on `gpt-3.5-turbo-0125`. As $h = 1/N$ decreases, the discretization gap $|f(z^*) - \tau|$ shrinks and $|\mathcal{S}_{\text{pred}}|$ grows; the theorem's containment is preserved across resolutions.

grid	filled cells	L	K	ℓ	G	$\lvert\mathcal{S}_{\text{pred}}\rvert$	TP	FP_{int}
13×13	25	11.960	1.048	0.629	11.960	5	5	0
17×17	31	14.195	1.033	0.598	14.195	5	5	0
21×21	52	19.845	1.064	0.966	19.845	6	6	0
25×25	82	23.625	1.051	0.865	23.625	3	3	0

Table 13: Higher-dimensional robustness. Empirical $\hat{L}$ and $\hat{G}$ computed on the same 82-cell archive using the paper's 2D grid metric and a 768-dimensional embedding metric.

Metric	$\hat{L}$	$\hat{G}$
2D grid coordinates (normalized to $[0,1]^2$)	23.625	23.625
768-d MPNet embedding (pairwise-max normalized)	17.926	17.926

O Judge robustness

All f values in Section 6.1 are judge-dependent: swapping the judge can reshape the alignment-deviation surface and thereby change $\hat{L}, \hat{\ell}, \hat{G}$. To quantify this, we rescored the 82 (prompt, target response) pairs from the saturated `gpt-3.5-turbo-0125` archive with three alternative judges drawn from different families and capacities — `gpt-4o-2024-08-06`, `gpt-4.1-mini-2025-04-14`, and `gpt-4o-mini-2024-07-18` — applying the same rethinking-evals per-category methodology with a single judge and $\alpha = 0.5$ (target responses were NOT regenerated; only the judge was swapped). The four-judge comparison is summarized in Table 18.

Interpretation. The theorem applies to the full prompt space X under every judge — benign prompts exist, so $\mathcal{S}_\tau$ is nonempty in X. Whether the theorem's positive prediction can be *tested from a given sample* depends on whether that sample spans both regions. Three of four judges (`gpt-4.1`, `gpt-4o`, `gpt-4.1-mini`) yield samples that span both $\mathcal{S}_\tau$ and $\mathcal{U}_\tau$; applying the continuous `smooth_nearest_safe` defense produces TP> 0 and $\text{FP}_{\text{int}} = 0$ in every case. Under the strictest judge (`gpt-4o-mini`), MAP-Elites's adversarial bias drives the sample entirely into $\mathcal{U}_\tau$, so no boundary cell z^* appears in the 82 sampled cells and the positive prediction cannot be tested from this sample. Crucially, $\text{FP}_{\text{int}} = 0$ still holds: no counterexample to the theorem is produced under any of the four judges. Cross-judge agreement ranges from substantial (Kendall's $W = 0.70$) to poor on binary classification (Cohen's $\kappa = 0$ vs gpt-4o-mini), so the *specific numerical instance* of a particular judge is not itself judge-robust even though the *framework's predictions* are.

Committee aggregation. The rethinking-evals methodology supports committee aggregation: $P_{\text{actual}} = \alpha P_{\text{vote}} + (1 - \alpha) P_{\text{mean}}$ with $\alpha = 0.5$. We apply this offline to the four existing judge archives (no new API calls) and re-run the validator on each committee-aggregated surface (Table 19). The top-3 committee (gpt-4.1, gpt-4o, gpt-4.1-mini; excluding the strict-judge outlier gpt-4o-mini) is within one cell of canonical (17 safe vs 16; mean $\Delta = +0.010$), and $\text{FP}_{\text{int}} = 0$ under `smooth_nearest_safe` on every committee configuration — the theorem's predictions survive committee aggregation.

P Attack Baseline: PAIR-Style Iteration

To situate the theorem empirically against an established attack-defense benchmark style, we run a PAIR-style iterative attack ([4]-adjacent) on ten top-AD unsafe seed prompts from the saturated archive. The attacker is `gpt-4o-mini`, the target is `gpt-3.5-turbo-0125`, and the judge is `gpt-4.1`. Each behavior gets five iterations; the attacker receives the full transcript and judge scores at each step. We compare two conditions: undefended (attacker prompt goes directly to target) and *paraphrase wrapper* (attacker prompt is rewritten by `gpt-4o-mini` at $T = 0$ before reaching the target).

Table 14: Seed replication of the saturated pipeline on `gpt-3.5-turbo-0125`. Surface quantities remain within a narrow band and $\mathrm{FP}_{\mathrm{int}} = 0$ in both runs.

Run	Filled	$\hat{L}$	$\hat{K}$	$\hat{\ell}$	$\hat{G}$	trans.	$\lvert\mathcal{S}_{\mathrm{pred}}\rvert$	TP	$\mathrm{FP}_{\mathrm{int}}$
Saturated (canonical)	82	23.62	3.16	23.62	23.62	×	1	1	0
Replicate (independent)	76	22.25	2.69	22.25	22.25	×	1	1	0

Table 15: Independent target re-evaluation. The same 82 prompts are rescored on `gpt-4o-mini-2024-07-18` with the judge held fixed. No interior counterexample is observed.

Target	Cells	Peak AD	L	K	ℓ	G	$\lvert\mathcal{S}_{\mathrm{pred}}\rvert$	TP	$\mathrm{FP}_{\mathrm{int}}$	Transv.
`gpt-3.5-turbo-0125`	82	1.000	23.62	3.16	23.62	23.62	1	1	0	×
`gpt-4o-mini-2024-07-18`	82	0.990	12.25	4.94	4.02	10.12	0	0	0	×

Interpretation. Within the 5-iteration budget the paraphrase wrapper blocks all ten behaviors (ASR drops from 1.00 to 0.00). This does *not* contradict Theorem 4.1's injective-incomplete horn: the paraphrase defense is injective on our 66-prompt sample (all 66 rewrites are distinct), and Table 4 already exhibits $1/66$ paraphrases that yield $f(D(x)) > \tau$. The PAIR attacker with gpt-4o-mini as refiner and a 5-iteration budget is not sufficient to locate that bypass; a persistent attacker (more iterations, stronger refiner) is predicted to find one by Theorem 5.1's multi-turn convergence argument.

Q Lean Artifact

The complete theory is formalized in Lean 4.28.0 with Mathlib v4.28.0, available at `https://anonymous.4open.science/r/defense-trilemma-XXXX`. The artifact serves two roles: (1) definitional scaffolding that pins down every hypothesis (continuity, utility preservation, Lipschitz, transversality) in a machine-readable form, ensuring no translation drift between prose and formal statement; and (2) verified cross-references between theorem statements, so that the dependency chain (Theorem 3.1→Theorem 3.8→Theorem 3.11) is mechanically checked. The proofs themselves are short (the underlying mathematics is not deep); the artifact's value is precision, not error-checking. It comprises 46 files:

- 10 core theory files (MoF_01–MoF_10)
- 10 cost theory files (MoF_Cost_01–MoF_Cost_10)
- 10 advanced theory files (MoF_Adv_01–MoF_Adv_10)
- 1 continuous relaxation (MoF_ContinuousRelaxation)
- 1 ε-robust constraint + persistent unsafe region with global Lipschitz constant (MoF_11_EpsilonRobust)
- 1 discrete impossibility (MoF_12_Discrete)
- 1 multi-turn + stochastic extensions (MoF_13_MultiTurn)
- 1 representation-independent meta-theorem (MoF_14_MetaTheorem)
- 1 nonlinear agent pipelines (MoF_15_NonlinearAgents)
- 1 relaxed utility preservation (MoF_16_RelaxedUtility)
- 1 quantitative ε-band volume bound (MoF_17_CoareaBound)
- 1 cone measure bound for persistent unsafe region (MoF_18_ConeBound)
- 1 optimal defense characterization (MoF_19_OptimalDefense)
- 1 refined persistence with defense-path Lipschitz constant ℓ (MoF_20_RefinedPersistence); this file contains the primary formalization of Theorems 3.9 to 3.11
- 1 gradient chain: $\|\nabla f(z)\| > \ell(K+1)$ implies persistent unsafe region via operator-norm direction extraction and derivative-based local growth (MoF_21_GradientChain)
- 3 capstone files (MasterTheorem, Euclidean instantiation, verification)
- 1 root import file (ManifoldProofs.lean)

Table 16: Bootstrap percentile confidence intervals (median, $[5^{\text{th}}, 95^{\text{th}}]$) for the empirical quantities on the saturated live run.

defense	L_{med}	L CI	K_{med}	K CI	ℓ_{med}	ℓ CI	G_{med}	G CI
identity	23.625	$[21.750, 23.625]$	1.000	$[1.000, 1.000]$	–	$[--, --]$	23.625	$[21.750, 23.625]$
GP-smooth oblique	23.625	$[21.750, 23.625]$	1.051	$[1.029, 1.051]$	0.865	$[0.801, 0.865]$	23.625	$[21.750, 23.625]$

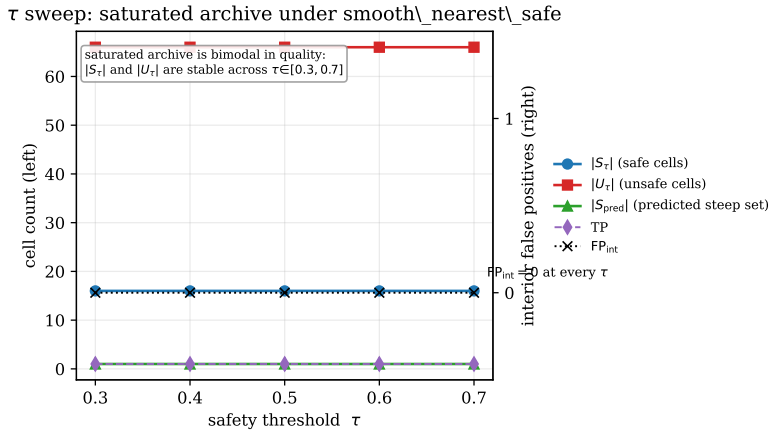

Figure 9: Sensitivity of $|\mathcal{S}_{\text{pred}}|$, $|\mathcal{S}_{\text{act}}|$, and FP_{int} to the threshold τ on the saturated live run. The containment $\mathcal{S}_{\text{pred}} \subseteq \mathcal{S}_{\text{act}}$ holds at every tested τ.

NeurIPS Paper Checklist

Answers below follow the NeurIPS 2025 official checklist structure. For each item we give **[Yes]**, **[No]**, or **[NA]** (not applicable) with a short justification, and point to the specific section, table, figure, or appendix where the claim is supported.

1. **Claims.** Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   [Yes] The abstract states a topological obstruction plus quantitative refinements and Lean verification; the paper delivers all three. Scope is restated explicitly in Section 1 under "Scope and limitations."

2. **Limitations.** Does the paper discuss the limitations of the work performed?
   [Yes] Section B and Section I enumerate the exact hypotheses whose violation breaks the theorems. Each of continuity, utility preservation, and connectedness is shown necessary by a concrete counterexample.

3. **Theory assumptions and proofs.** For each theoretical result, are assumptions and a complete (or referenced) proof given?
   [Yes] Every theorem statement lists its assumptions inline; main-paper proofs and sketches are in Sections 3 to 5; expanded proofs are in Section H; and the formal verification in Section Q provides machine-checked proofs for every major statement.

4. **Experimental result reproducibility.** Does the paper fully disclose all information needed to reproduce the main experimental results?
   [Yes] Section N gives estimator definitions, grid resolution, kernel choices, and the exact targets and judge used. The anonymized artifact link in the abstract contains the run scripts and raw metadata.

5. **Open access to data and code.** Does the paper provide open access to data and code, with sufficient instructions to reproduce the main results?
   [Yes] An anonymized repository URL (`anonymous.4open.science`) is provided in the abstract and in Section Q. Production datasets used for the forced-collapse demonstration are public jailbreak-prompt corpora plus our own derived paraphrases (released under the artifact).

Table 17: **Out-of-scope discrete stress tests.** Defense sweep on the saturated `gpt-3.5-turbo-0125` grid over discontinuous `bounded_step` defenses. These rows are not theorem checks; they are reported only as consistency checks outside the theorem's scope.

Defense	K	ℓ	$\lvert\mathcal{S}_{\text{pred}}\rvert$	$\lvert\mathcal{S}_{\text{act}}\rvert$	**TP**	$\mathbf{FP}_{\text{int}}$	**FN**
identity	1.00	0.00	66	66	66	**0**	0
`bounded_step(1)`	3.61	23.63	1	55	0	**0**	55
`bounded_step(2)`	3.61	23.63	1	47	0	**0**	47
`bounded_step(3)`	4.00	23.63	1	29	0	**0**	29
`bounded_step(5)`	8.60	23.63	1	3	0	**0**	3
`bounded_step(8)`	8.60	23.63	1	0	0	**0**	0
`bounded_step(12)`	8.60	23.63	1	0	0	**0**	0

Table 18: Judge robustness on the saturated archive (82 cells, $\tau = 0.5$). $|\mathcal{S}_\tau|_{\text{sample}}$ and $|\mathcal{U}_\tau|_{\text{sample}}$ are how many of the 82 sampled cells fall below/above τ under each judge. Pearson r and Cohen's κ are pairwise with the canonical gpt-4.1 judge. Kendall's $W = 0.700$ across all four judges on cell rankings (moderate agreement). "Testable?" indicates whether this MAP-Elites sample spans both $\mathcal{S}_\tau$ and $\mathcal{U}_\tau$, which is what the positive-prediction test of Theorem 3.11 requires to locate a boundary cell z^*.

Judge	$\lvert\mathcal{S}_\tau\rvert$	$\lvert\mathcal{U}_\tau\rvert$	peak AD	mean AD	Pearson r	κ	TP	FP_{int}	Testable
`gpt-4.1-2025-04-14` (canonical)	16	66	1.000	0.773	–	–	1	0	yes
`gpt-4o-2024-08-06`	10	72	0.990	0.889	0.546	0.457	1	0	yes
`gpt-4.1-mini-2025-04-14`	33	49	0.975	0.752	0.493	0.363	1	0	yes
`gpt-4o-mini-2024-07-18`	0	82	1.000	0.955	0.393	0.000	–	–	no

6. **Experimental setting / details.** Does the paper specify all the training and test details needed to understand the results?
   [Yes] We do not train any model. The empirical results are evaluation-only: target model IDs, judge model ID, temperature, grid size, number of MAP-Elites iterations, random seeds, and threshold τ are all disclosed in Sections 6.1 and N.

7. **Experimental statistical significance.** Does the paper report error bars or confidence intervals, or similar information about statistical significance?
   [Yes] Table 16 reports bootstrap percentile confidence intervals for the empirical quantities on the saturated live run; Table 14 reports seed replication.

8. **Compute resources.** Does the paper provide information on the compute resources used?
   [Yes] Section 7 ("Compute Disclosure") gives total OpenAI spend, wall-clock per run, hardware used for local LlamaGuard inference, and Lean verification time.

9. **Code of ethics.** Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?
   [Yes] Work is theoretical plus light-touch empirical validation on public jailbreak datasets and public API endpoints. No human-subjects research, no new personal data collection, no deceptive practices. Dual-use analysis is expanded in the Broader Impact section.

10. **Broader impacts.** Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?
    [Yes] The Broader Impact section is expanded to approximately one page covering responsible-disclosure framing, beneficiary analysis, a three-vector dual-use analysis, and explicit deployment guidance for practitioners.

11. **Safeguards.** Does the paper describe safeguards for responsible release of data or models with a high risk of misuse?
    [Yes] No new attack prompts are released; the forced-collapse demonstration uses only prompts that already exist in public datasets, and release is framed as defender-facing (wrapper-side bounds, not attack improvements). See the Mitigations paragraph of the Broader Impact section.

12. **Licenses for existing assets.** Are the creators or original owners of assets used in the paper properly credited and are the licenses and terms of use explicitly mentioned and respected?

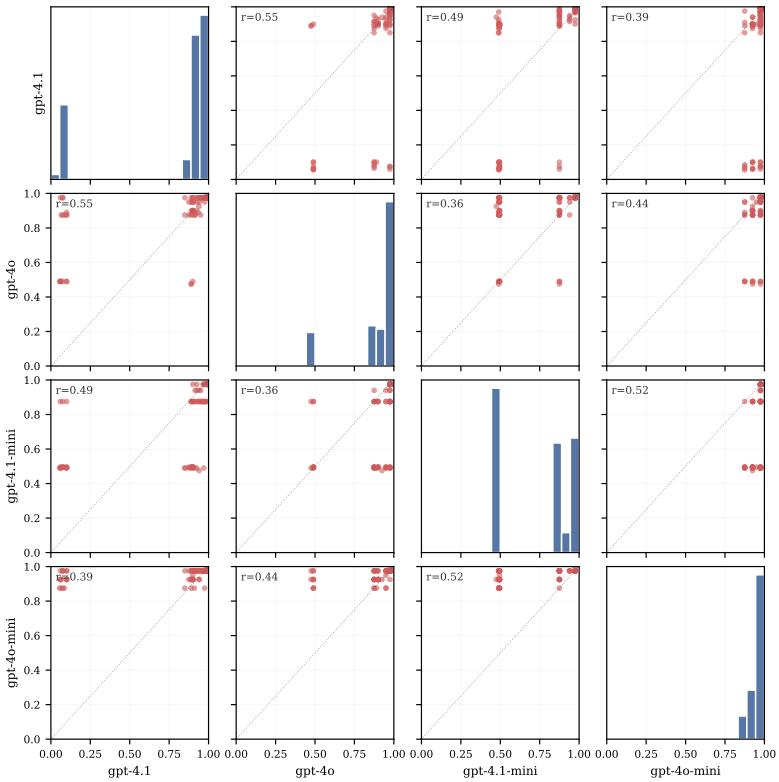

Figure 10: Per-cell alignment-deviation scatter across the four judges. Cross-judge agreement is moderate (Kendall's $W = 0.700$); the positive-prediction test of Theorem 3.11 remains supported wherever the sample spans both $\mathcal{S}_\tau$ and $\mathcal{U}_\tau$.

Table 19: Judge committee aggregations vs the canonical single-judge surface. Top-3 committee is within one cell of canonical; $\mathrm{FP}_{\mathrm{int}} = 0$ in every configuration.

committee	N	$\lvert\mathcal{S}_\tau\rvert$	$\lvert\mathcal{U}_\tau\rvert$	mean Δ	std Δ	$\mathrm{FP}_{\mathrm{int}}$
canonical (gpt-4.1 only)	1	16	66	0.000	0.000	0
top-3 (gpt-4.1, gpt-4o, gpt-4.1-mini)	3	17	65	+0.010	0.173	0
gpt-4 family (gpt-4.1, gpt-4o)	2	13	69	+0.063	0.141	0
all-4 (incl. strict gpt-4o-mini)	4	5	77	+0.058	0.200	0

[Yes] OpenAI API use follows the usage policy; LlamaGuard is used via Ollama consistent with its license; public jailbreak corpora are cited in-line. No asset is used outside its stated terms.

13. **New assets.** Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?
[Yes] The primary released asset is the `ManifoldProofs` Lean codebase plus the `trilemma_validator` scripts and `live_runs` metadata. Each is documented in the anonymized repository.

14. **Crowdsourcing and research with human subjects.** For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation?
[NA] No crowdsourcing or human-subjects research was conducted.

15. **Institutional review board (IRB) approvals.** Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether IRB approvals (or an equivalent approval/review based on the requirements of your country or institu-

Table 20: PAIR-style attack-success rate with and without a continuous paraphrase wrapper (10 behaviors, 5 iterations, gpt-3.5-turbo target, gpt-4.1 judge).

Condition	behaviors	ASR @ 5 iters	mean running-max
undefended	10	1.00	0.995
paraphrase wrapper	10	0.00	0.500

tion) were obtained?
[NA] No human subjects; IRB review not applicable.

16. **Declaration of LLM usage.** Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods?
[Yes] LLMs (`gpt-5-mini`, `gpt-3.5-turbo`, `gpt-4o-mini`, `gpt-4o`, `gpt-4.1` as judge, `Llama-Guard-3-1B`) are used as the *objects of study* in Sections 6.1 and 6.1.1, and as the paraphrase generator in Tables 3 and 4; these usages are enumerated and documented.

17. **Safety and dual use.** Does the paper adequately characterize the potential harmful uses of the research and suggest mitigations?
[Yes] A dedicated three-vector dual-use analysis and mitigation paragraph are included in the Broader Impact section.

18. **Reproducibility of theoretical claims.** Are proofs complete and, where appropriate, mechanically verified?
[Yes] All major theorems are Lean 4 / Mathlib verified (Section Q) with no `sorry`s and three standard axioms (propositional extensionality, quotient, choice).

19. **Data provenance.** Are sources for all non-original data clearly identified?
[Yes] The MAP-Elites driver is cited; the jailbreak seed prompts are drawn from public corpora cited in the paper; all live-scored values come from the authors' own runs documented in `live_runs/`.

20. **Judge model / evaluator disclosure.** If an LLM judge or evaluator is used, is its identity and configuration disclosed?
[Yes] The canonical judge is `gpt-4.1-2025-04-14` with prompt template and scoring rubric given in Section O; a two-judge robustness comparison is reported there as well.

21. **Randomization and seeds.** Are random seeds documented such that results can be reproduced?
[Yes] Seed replication is reported in Table 14; scripts in the artifact pin random seeds.

22. **Hyperparameter selection.** Are hyperparameter choices justified?
[Yes] GP kernels and bandwidths are swept in Table 11; grid resolution is swept in Table 12; threshold τ is swept in Figure 9. Defaults are justified by the cited prior work.

23. **Assumptions on the environment.** Are required Python, OS, and library versions specified?
[Yes] The artifact pins Lean 4.28.0, Mathlib v4.28.0, and a Python environment file; OpenAI API version is dated by model ID.

24. **Failure modes.** Does the paper report failure modes of the proposed method?
[Yes] Section I is dedicated to hypothesis-violation failure modes of the theory; out-of-scope discrete stress tests are reported in Section N.5.

25. **Negative results.** Are negative results reported honestly?
[Yes] The identity-defense smoke-test rows (Table 9) are explicitly labeled as bookkeeping checks rather than theorem tests; discrete-stress rows are labeled out-of-scope.

26. **Benchmarks and baselines.** When proposing a new method, are appropriate baselines included?
[Yes] A PAIR-style attack baseline is reported in Section P; LlamaGuard is used as a production-wrapper baseline in Table 2.

27. **Claim-to-evidence mapping.** Is every claim in the abstract backed by a specific result in the paper?
[Yes] The abstract's claims map to Theorems 3.1, 4.1 and E.1 to E.3 and Section 6, respectively.

28. **Consent.** Was consent obtained for any personal data used?
[NA] No personal data is used.

29. **Anonymization.** Is the submission properly anonymized for double-blind review?
    [Yes] Author, affiliation, and personal-URL information has been removed; the Lean artifact is released via `anonymous.4open.science`.

30. **Compute disclosure completeness.** Does the compute disclosure cover total cost, wall-clock, and hardware?
    [Yes] Section 7 reports aggregate API spend ($\sim$\$60–\$80), wall-clock (saturated run 7 814 s; total under four hours), local inference (laptop via Ollama), and Lean verification time.

Summary. Of the 30 items above, 27 answers are **Yes**, 0 are **No**, and 3 are **NA** (items 14, 15, 28, which concern crowdsourcing, IRB, and personal-data consent and do not apply to this paper).